🏦 Bank Customer Churn Analysis Using Pandas, Matplotlib, and Seaborn

📖 Project Overview

Customer churn refers to customers who stop using a bank's services. Retaining existing customers is generally more cost-effective than acquiring new ones. This project analyzes customer data to identify patterns and factors contributing to customer churn.

Using Pandas, Matplotlib, and Seaborn, we perform data cleaning, exploratory data analysis (EDA), and visualization to uncover actionable business insights.

🛠 Tools and Libraries Used

Tool	**Purpose**
🐼 Pandas	Data manipulation and analysis
🔢 NumPy	Numerical operations
📊 Matplotlib	Data visualization
🎨 Seaborn	Statistical visualization
📓 Jupyter Notebook	Development environment

🗂 Dataset Description

The dataset contains customer information from a bank.

Column Name	Description
CustomerId	Unique customer identifier
Surname	Customer surname
CreditScore	Credit score of customer
Geography	Customer country
Gender	Male/Female
Age	Customer age
Tenure	Number of years with bank
Balance	Account balance
NumOfProducts	Number of bank products
HasCrCard	Credit card ownership
IsActiveMember	Active membership status
EstimatedSalary	Estimated annual salary
Exited	Churn status (1 = Churned, 0 = Retained)

🔄 Steps Followed

1️⃣ 📥 Data Collection

- Obtained the Bank Customer Churn dataset.
- Loaded the dataset into a Pandas DataFrame.

2️⃣ 🔍 Data Understanding

- Examined dataset structure, dimensions, and data types.
- Reviewed column descriptions and business context.

3️⃣ 🧹 Data Cleaning

- Checked for missing values and duplicates.
- Verified data quality and consistency.

4️⃣ 📊 Exploratory Data Analysis (EDA)

- Performed Univariate Analysis.
- Performed Bivariate Analysis.
- Performed Multivariate Analysis.
- Generated statistical summaries.

5️⃣ 📈 Data Visualization

- Created bar charts, histograms, pie charts, box plots, scatter plots, and heatmaps.
- Used subplots to compare multiple variables effectively.

6️⃣ 📋 GroupBy & Pivot Table Analysis

- Analyzed customer behavior using aggregation techniques.
- Summarized key metrics across customer segments.

7️⃣ 🔥 Correlation Analysis

- Examined relationships between numerical features.
- Identified variables strongly associated with churn.

8️⃣ 💡 Business Insights Generation

- Identified customer groups with higher churn probability.
- Discovered patterns related to age, geography, balance, and activity status.

9️⃣ ✅ Recommendations

- Suggested customer retention strategies.
- Proposed targeted engagement and loyalty initiatives.

🔟 📝 Documentation & Reporting

- Compiled findings, visualizations, insights, and recommendations into a structured report.

💡 Key Business Insights

👥 Customer Demographics

- Customers aged above 45 show higher churn rates.
- Female customers churn slightly more frequently.

🌍 Geographic Trends

- Germany has the highest churn rate.
- France shows the highest retention rate.

💰 Financial Behavior

- Customers with higher balances are more likely to churn.
- Credit score has limited impact on churn.

🏦 Banking Activity

- Inactive members have significantly higher churn rates.
- Customers with multiple products tend to remain loyal.

📊 Visualizations Created

📉 Bar Charts

Used for:

- Customer Churn Distribution
- Churn by Gender

Purpose:

Compare categorical variables.

🥧 Pie Chart

Used for:

- Gender Distribution

Purpose:

Display proportion of male vs female customers.

📈 Histogram

Used for:

- Age Distribution

Purpose:

Understand distribution patterns.

📦 Box Plots

Used for:

- IsActive vs Churn

Purpose:

Identify outliers and spread of data.

🔵 Scatter Plot

Used for:

- Age vs Balance

Purpose:
Study relationships between numerical variables.

🔥 Heatmap

Used for:

- Correlation Analysis

Purpose:
Identify relationships among numerical features.

📊 Count Plot

Used for:

- Geography vs Churn

Purpose:
Displays the frequency of categorical variables.

📈 Line Chart

Used for:

- Average Balance vs Tenure

Purpose:
Helps identify increasing or decreasing relationships.

🎻 Violin Plot

Used For

- Age Distribution by Gender

Purpose:
Provides deeper insight into the spread of values.

🧩 Pair Plot

Used For

- Age, Balance, Estimated Salary, and Churn Analysis

Purpose:
Displays pairwise relationships among multiple numerical variables.

📑 Subplots

Used For

- Display multiple charts in a single figure for easier comparison.

Purpose:
Improved dashboard appearance and better comparison across variables.

⚡ Interactive Plotly Visualization

Used For

- Interactive Age vs Balance Analysis

Purpose:

Allows zooming, filtering, and hovering over data points. Also, enhances user interaction with visualizations.

📁 Files Included

The project consists of the following files and folders:

📈 Bank_Churn.csv – Original dataset used for customer churn analysis.

📊 Bank Customer Churn Analysis.ipynb - Jupyter Notebook containing data cleaning, EDA, visualizations, and analysis.

📜 README.md - Project overview, setup instructions, and documentation.

📄 Bank Customer Churn Analysis Project Doc - Project Documentation.

🚀 How to Use

1. Open Bank Customer Churn Analysis.ipynb using Jupyter Notebook or JupyterLab.
2. Run the notebook cells step by step to view data cleaning, analysis, and visualizations.
3. Modify the code to explore additional insights if needed.

✅ Recommendations

🎯 Improve Customer Engagement

- Target inactive customers with loyalty programs.
- Increase personalized communication.

🎁 Product Bundling

- Encourage customers to use multiple banking products.

🌍 Region-Specific Strategies

- Focus retention campaigns in Germany.

🧑‍💼 High-Value Customer Retention

- Monitor high-balance customers for early churn indicators.

📚 Conclusion

This Bank Customer Churn Analysis project provides a comprehensive understanding of customer behavior and churn patterns. By leveraging Python's data analysis and visualization libraries, banks can make data-driven decisions to improve customer retention, increase customer satisfaction, and enhance long-term profitability.

👤 Author

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